Quiz 2

The default gateway IP address for the remote router connected via the red port depends on the specific lab configuration and instructions provided in your course materials. However, based on common networking setups:

- **192.168.1.1** is a standard default gateway for many home routers.
- **172.16.255.254** is within the private IP range 172.16.0.0/12 and often used in corporate or lab environments.
- **10.10.10.254** is within the private IP range 10.0.0.0/8, which is also frequently used in custom lab environments.
- **169.16.1.254** is unusual because it does not fall into any of the reserved private IP ranges.

For a lab setup, **172.16.255.254** or **10.10.10.254** are more likely candidates, but the exact answer should match your specific lab manual or instructions. Check your Lab 2 documentation for confirmation.

The two end devices are:

1. **Laptop**
2. **Printer**

Explanation:

- **End devices** are devices that send or receive data within a network. Examples include laptops, printers, desktops, and phones.
- **Routers** and **switches** are **intermediary devices** that facilitate data flow between end devices, so they are not considered end devices.

The correct answer is:

Four echo requests, four echo replies

Explanation:

When you ping a device, the default behavior of most operating systems (such as Windows, Linux, or macOS) is to send **four ICMP echo requests** and expect **four ICMP echo replies** in return. A successful ping indicates that all four requests received replies, confirming connectivity to the default gateway.

The correct answers are:

Router
Switches

Explanation:

- **Intermediary devices** are responsible for managing and directing data traffic within a network. Examples include routers, switches, firewalls, and access points.
- **Laptop** and **Printer** are **end devices** because they are the sources or destinations of data within a network.

The correct answer is:

DNS (Domain Name System)

Explanation:

- **DNS** resolves human-readable URLs (e.g., www.example.com) to their corresponding IP addresses, enabling devices to locate servers and communicate over the internet.
- **Router** directs data packets between networks.
- **DHCP (Dynamic Host Configuration Protocol)** assigns IP addresses dynamically to devices in a network.
- **FTP (File Transfer Protocol)** is used for transferring files between systems, not for resolving URLs.

True

Explanation:

Packet switching is a method used in networking where data is broken into smaller packets. These packets are then routed independently through the network and could take different paths to reach the destination. Upon arrival, the packets are reassembled in the correct order.

The correct answers are:

Copper
Fiber
Wireless

Explanation:

- **Copper** is a traditional wired medium used for Ethernet connections.
- **Fiber** is a high-speed wired medium that uses optical cables for data transmission.
- **Wireless** refers to communication over airwaves, such as Wi-Fi, Bluetooth, or cellular networks.
- **SSD** (Solid-State Drive) is a storage medium, not a type of network transmission medium.

The correct answer is:

DHCP (Dynamic Host Configuration Protocol)

Explanation:

- **DHCP** automatically provides IP addresses to devices on a network, along with other network configuration settings.
- **Switch** and **Hub** are networking devices used for managing and connecting devices but do not automatically assign IP addresses.
- **RJ45** is a type of connector, not a device used for IP assignment.

The correct answer is:

Because of EMI (Electromagnetic Interference)

Explanation:

- UTP (Unshielded Twisted Pair) cables have twisted wires to reduce electromagnetic interference (EMI) and crosstalk, which can degrade signal quality and performance. The twisting helps to balance the signal between the pairs, minimizing interference from external sources.

The correct answer is:

MAC address

Explanation:

- The **Data Link Layer** in the OSI model deals primarily with MAC (Media Access Control) addresses, which are unique hardware addresses used to identify devices on a local network.
- IP addresses are handled at the **Network Layer**, not the Data Link Layer.
- "Home address" is not a standard term used in relation to the OSI model.

The correct answer is:

ipconfig /all

Explanation:

- The `**ipconfig /all**` command displays detailed information about your network configuration, including the physical (MAC) address of your NIC (Network Interface Card).
- Other options like `**ipconfig**` show basic network information, but `**ipconfig /all**` provides the full details.
- The other options (`**ipconfig /more**`, `**ipconfig /NIC**`, and `**TF**`) are not valid commands for displaying the physical address of a NIC.

The correct answer is:

Protocol Data Unit (PDU)

Explanation:

- A **Protocol Data Unit (PDU)** is a general term used to describe data at any layer of a networking model (e.g., frame at the Data Link Layer, packet at the Network Layer, segment at the Transport Layer).
- **Frame**, **Packet**, and **Segment** are specific types of PDUs used at different layers of the OSI model.

The correct answer is:

All the above

Explanation:

- Ethernet can use various physical topologies such as bus, star, ring, and mesh.
- Ethernet networks use a broadcast signal to transmit data to all devices on the network.
- Ethernet utilizes hardware such as hubs, switches, RJ45 connectors, and sometimes coaxial cables, depending on the network setup.

The correct answers are:

Local Area Network (LAN)
Wide Area Network (WAN)

Explanation:

- **LAN** (Local Area Network) covers a small geographic area, such as an office or home network.
- **WAN** (Wide Area Network) spans a larger geographical area, connecting multiple LANs across regions or even globally.
- **Peer-to-Peer** and **Client/Server** are network models, not network infrastructures.

The correct answer is:

Crossover cable

Explanation:

In a lab environment when connecting two routers (e.g., R1 and R2), a **Crossover cable** is used to directly connect the devices as it allows the sending and receiving pairs to be crossed for direct communication between two similar devices.

- **Straight-through cables** are used for connections between devices of different types (e.g., PC to switch or router to switch).
- **Coaxial cable** is used in older networking technologies, like Ethernet over coax, but not typically for connecting routers directly.
- **Serial cable** is used for connecting networking devices with serial interfaces, not typically routers in a LAN setup.

The correct answer is:

3C-58-C2-F5-88-95

Explanation:

- **3C-58-C2-F5-88-95** is a valid MAC (Media Access Control) address, which is a unique identifier assigned to network interfaces.
- **12345** is not a valid MAC address format.
- **192.168.5.17** is an IP address.
- **2+3+D** is not a valid MAC address.

The correct answer is:

10 base T

Explanation:

- **10BaseT** is another name for **Star Topology** in Ethernet networks, where devices are connected to a central hub or switch in a star configuration.
- The other options (**10 base 2**, **10 base 5**, and **10 band two**) are not standard network topology terms.

The correct answer is:

Internet Assigned Numbers Authority (IANA)

Explanation:

- **IANA** is responsible for managing and overseeing the global allocation of IP addresses, domain names, and other Internet resources.
- **Internet Society (ISOC)** focuses on the overall health and progress of the internet.
- **Internet Architecture Board (IAB)** focuses on technical development and the architecture of the Internet.
- **Internet Research Task Force (IRTF)** conducts long-term research into the technical, architectural, and operational aspects of the internet.

The correct answer is:

Router

Explanation:

- A **Router** operates at **Layer 3 (Network Layer)** of the OSI model. It is responsible for routing data packets between different networks based on IP addresses.
- A **Switch** primarily operates at **Layer 2 (Data Link Layer)** but can also work at Layer 3 in the case of Layer 3 switches.
- A **Hub** is a simple device that operates at **Layer 1 (Physical Layer)**.
- A **PC/Laptop** is an end device and does not function at a specific OSI layer in terms of forwarding network traffic.

The correct answer is:

Straight-through cable

Explanation:

In a typical lab setup like "Intro to PT," a **Straight-through cable** is used to connect different types of devices, such as an Eagle server (end device) to an R1-ISP router (networking device).

- **Crossover cable** is used to connect similar devices, such as two routers or two switches.
- **Roll-over cable** is used for console access to configure networking devices.
- **Coaxial cable** is not commonly used in modern Ethernet networking.

True

Explanation:

Network bandwidth is the amount of data that can be transmitted over a network connection in a specific amount of time, typically measured in **bits per second (bps)** or its multiples (e.g., kbps, Mbps, Gbps). This measurement reflects the capacity of the network to handle data traffic.

The correct answers are:

Data
Voice
Video

Explanation:

- **Converged networks** are designed to integrate multiple forms of communication—such as data (emails, files), voice (VoIP calls), and video (streaming, video conferencing)—into a single network infrastructure.
- **Hard drive** is a storage device and not something delivered by a network.

The correct answer is:

Packet-switched network.

Explanation:

- A **packet-switched network** is considered **connectionless** because data is broken into packets, and each packet is routed independently, possibly taking different paths to reach the destination. There is no need to establish a dedicated connection beforehand.
- A **circuit-switched network** is **connection-oriented**, as it requires a dedicated path to be established between the sender and receiver before data transmission (e.g., traditional telephone systems).
- **Packet printing network** is not a standard term in networking.

The correct answer is:

ICMP (Internet Control Message Protocol)

Explanation:

- **ICMP** is the protocol used to send echo requests and replies, as seen in the **ping** command. It helps in testing connectivity and diagnosing network issues.
- **HTTP (HyperText Transfer Protocol)** is used for web traffic.
- **DNS (Domain Name System)** resolves domain names to IP addresses.
- **FTP (File Transfer Protocol)** is used for transferring files between systems.

The correct answer is:

FF-FF-FF-FF-FF-FF

Explanation:

- **FF-FF-FF-FF-FF-FF** is the MAC address used for broadcasting at the Data Link Layer. It indicates that the message is intended for all devices on the local network segment.
- **192.168.2.17** and **172.164.24.9** are IP addresses, not broadcast signals.
- **FF-FF-FF-CD-FF-FF** is not a valid broadcast MAC address because it deviates from the standard broadcast format.

The correct answer is:

Straight-through cable

Explanation:

In Lab 2, when connecting a laptop to a Linksys router, a **Straight-through cable** is used. This type of cable is commonly employed to connect different types of devices, such as a PC (or laptop) to a router or switch.

- **Rollover cable** is used for console connections to configure devices.
- **Cross-over cable** is used for connecting similar devices, like two PCs or two routers.
- **Coaxial cable** is not typically used in Ethernet-based network setups.

The correct answers are:

Expensive
Needs an administrator.

Explanation:

- **Expensive**: Client/server networks require dedicated servers, specialized software, and additional hardware, which can be costly to set up and maintain.
- **Needs an administrator**: These networks require skilled administrators to manage the servers, user accounts, security, and troubleshooting.
- **Expansion allowed** is an **advantage** of client/server networks, not a disadvantage.
- **All the above** is incorrect because "Expansion allowed" does not describe a disadvantage.

The correct answer is:

Peer to Peer

Explanation:

- In a **Peer-to-Peer (P2P)** network, devices act as both clients and servers simultaneously. Each device can request services from other devices (acting as a client) and provide services to other devices (acting as a server).
- **Client/server** networks have dedicated roles for clients and servers, so devices do not act as both.
- **Workstation to printer** is not a type of network; it describes a specific connection for printing.
- **None of the above** is incorrect because Peer-to-Peer fits the description.

The correct answer is:

WAN network

Explanation:

- **WAN network** (Wide Area Network) is an example of an **extranet network** because it connects multiple different organizations or locations over a wide geographical area.
- **Hospital**, **Local Office**, and **SAN network** (Storage Area Network) are more specific to single or closely integrated environments, not generally involving external access.

The correct answers are:

Fault Tolerance
Scalability
Quality of Service (QoS)
Security

Explanation:

- **Fault Tolerance** ensures the network remains operational even when failures occur.
- **Scalability** refers to the ability of a network to grow and handle increased traffic or more devices.
- **Quality of Service (QoS)** manages network traffic to prioritize certain types of data or applications.
- **Security** encompasses measures to protect network data from unauthorized access, attacks, and breaches.

The correct answer is:

Bit per second

Explanation:

Network bandwidth is measured in the number of bits that can be transmitted in a single second, which is referred to as **bits per second (bps)**.

The correct answers are:

Texting
Blog
Podcasting

Explanation:

- **Texting** is a form of direct communication.
- **Blog** provides written content that communicates ideas or information.
- **Podcasting** involves audio content, which communicates messages or stories to an audience.
- **Wikis** can be a form of collaborative communication, but the others are more widely recognized individual communication methods.

The correct answers are:

Host
End devices

Explanation:

- **Host** refers to any device connected to a network that can send or receive data.
- **End devices** are the final devices in the network that interact with the network, such as computers, smartphones, printers, and servers.
- **Printer** and **Server** are specific types of devices, but not the general term used to describe all devices on a network.

True

Explanation:

An **Extranet** allows external organizations, partners, or stakeholders to securely access certain parts of an organization's data, such as applications, documents, or resources, while maintaining controlled access and ensuring security.

The correct answer is:

Internet Control Message Protocol

Explanation:

ICMP (Internet Control Message Protocol) is used for managing errors and providing information about network conditions, such as handling error messages, ping requests, and other network diagnostics.

The correct answer is:

Routers

Explanation:

- The **Default Gateway** is a configuration primarily set on **routers**, allowing them to manage communication between a local network and external networks, such as the internet.
- **Hubs**, **Printers**, and **Motherboards** do not typically have a default gateway configured.

The correct answer is:

Electromagnetic Interference (EMI)

Explanation:

EMI refers to the interference caused by electromagnetic waves that can disrupt electronic communication and cause signal degradation in networks and devices.

The correct answer is:

DHCP

Explanation:

- **DHCP (Dynamic Host Configuration Protocol)** automatically assigns IP addresses to devices on a network, eliminating the need for manual configuration.
- **Switch**, **Hub**, and **RJ45** do not provide IP addresses automatically.

The correct answer is:

Router

Explanation:

- A **Router** operates at **Layer 3 (Network Layer)** of the OSI model, responsible for forwarding data packets between different networks based on IP addresses.
- **Switch** primarily operates at **Layer 2 (Data Link Layer)**, but some advanced Layer 3 switches can handle network routing tasks.
- **Hub** operates at **Layer 1 (Physical Layer)**, and **PC Laptop** is an end device, not a networking device.

The correct answer is:

MAC address

Explanation:

- The **Data Link Layer** in the OSI model deals primarily with **MAC addresses**, which are unique identifiers assigned to network interfaces.
- **IP addresses** are handled at the **Network Layer** (Layer 3).
- **Home address** is not directly associated with the Data Link Layer.

The correct answer is:

Routers

Explanation:

- The **Default Gateway** is primarily configured on **Routers**. Routers manage traffic between different networks and provide the necessary gateway for devices to access external networks like the internet.
- **Hubs**, **Printers**, and **Motherboards** do not typically have a default gateway configured.

The correct answer is:

Bit per second

Explanation:

Network bandwidth is measured in bits per second (**bps**), which indicates how much data can be transmitted in one second over a network connection.

The correct answer is:

ICMP (Internet Control Message Protocol)

Explanation:

- In Lab 2, **ICMP** is the protocol used to send echo requests and receive echo replies, which are commonly used in the **ping** command to test network connectivity.
- **HTTP**, **DNS**, and **FTP** are application layer protocols and do not handle echo requests directly like ICMP does.

The correct answer is:

Hub

Explanation:

- A **Hub** operates at **Layer 1 (Physical Layer)** of the OSI model, as it simply replicates and broadcasts data to all connected devices without inspecting the content.
- **Router**, **Switch**, and **Bridge** operate at higher layers (Layer 3 and Layer 2, respectively).

The correct answer is:

Protocol Data Unit (PDU)

Explanation:

- **PDU** is the general term used to describe a piece of data at any layer of a networking model.
- A **frame** is used in the Data Link Layer,
- A **packet** is used in the Network Layer,
- A **segment** is used in the Transport Layer.

True

Explanation:

Network bandwidth is measured in bits per second (bps), which indicates the amount of data that can be transmitted in one second over a network connection.

True

Explanation:

In packet-switched networks, data is divided into packets, which can take different routes to reach the destination. Each packet is then reassembled in the correct order at the destination. This allows efficient use of network resources and better handling of network traffic.